

SUGGESTED SOLUTIONS TO 2021 H2 PHYSICS PRELIM PAPER 1

Q1

1 mol of water molecules has a mass of 0.018 kg.

Mtd1:

Number of molecules is

$$N = \frac{M}{M_m} N_A = \frac{0.200}{0.018} (6.02 \times 10^{23})$$

$$N \approx 6.7 \times 10^{24} \approx 10^{25}$$

Mtd2:

Water is H_2O , so the mass number is $16+1+1 = 18$.

$$N = \frac{M}{M_{\text{watermolecule}}} = \frac{0.200}{18u} = \frac{0.200}{18 \times 1.66 \times 10^{-27}} = 6.69 \times 10^{24} \approx 10^{25}$$

Ans: C

Q2